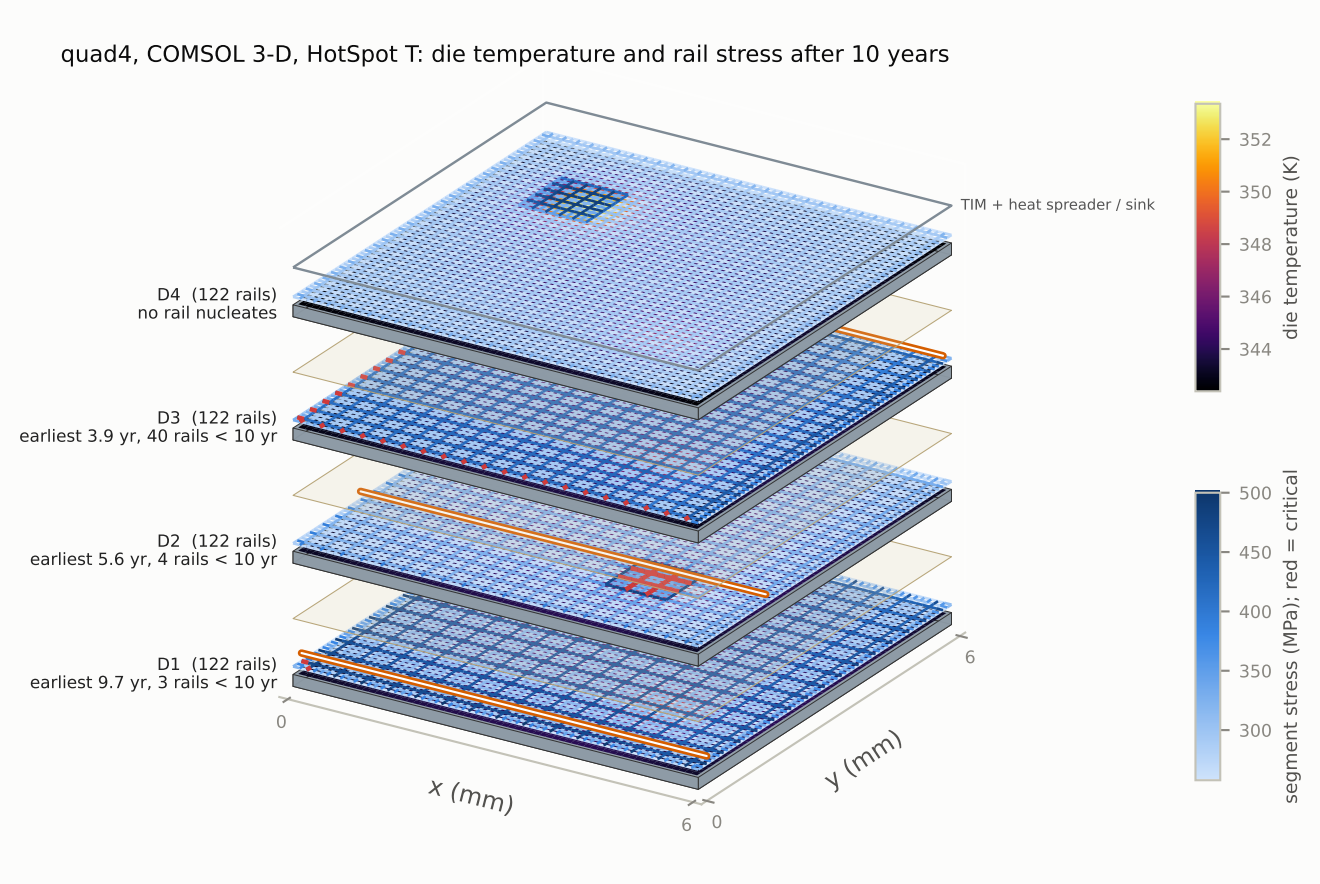

quad4, COMSOL 3-D, HotSpot T: die temperature and rail stress after 10 years
TIM + heat spreader / sink
D4 (122 rails)
no rail nucleates
D3 (122 rails)
earliest 3.9 yr, 40 rails < 10 yr
D2 (122 rails)
earliest 5.6 yr, 4 rails < 10 yr
D1 (122 rails)
earliest 9.7 yr, 3 rails < 10 yr
0
x (mm)
6
0
y (mm)
6
352
350
348
346
344
die temperature (K)
500
450
400
350
300
segment stress (MPa); red = critical